

Principles of Database Systems (CS307)

Zhong-Qiu Wang

Department of Computer Science and Engineering
Southern University of Science and Technology

- Most contents are from slides made by Stéphane Faroult and the authors of Database System Concepts (7th Edition).
- Their original slides have been modified to adapt to the schedule of CS307 at SUSTech.
- The slides are largely based on the slides provided by Dr. Yuxin Ma

Instructor

- **Dr. Zhong-Qiu Wang (王中秋)**
 - Associate Professor in CSE
 - Office: Room 517, South Tower, Engineering Building
 - Homepage: <https://zqwang7.github.io/>
 - Email: wangzq3@sustech.edu.cn
- **Background**
 - Postdoc, Carnegie Mellon University, 2021/09 ~ 2024/07
 - Research Scientist, Mitsubishi Electric Research Lab, 2020/06 ~ 2021/08
 - Ph.D. student, Ohio State University, 2013/08 ~ 2020/05
 - Undergrad, Harbin Institute of Technology, 2009/08 ~ 2013/07
- **Research interests: AI + Speech and audio processing**
 - Perception, understanding, and generation of speech and audio signals
- **Office Hour: [Mondays 4pm-6pm @ my office](#)**

Lecture Notes, Blackboard and WeCom Group

- There is a [Blackboard](#) site for this course
 - Lecture slides, Lab sheets, Assignments, Project requirements, Announcements
 - Will **upload** lecture slide before the lecture
- Teaching Laboratory Technician (教辅)
 - Yueming Zhu (朱悦铭老师)
- Lab sessions
 - Group 1: Mon 9-10 (Zhong-Qiu Wang)
 - Group 2: Mon 7-8 (Yueming Zhu)
 - Group 3: Wed 3-4 (Yueming Zhu)
- 3 TAs, one for each lab session
 - 黄亮、段晟俊、宋一鸣

Teaching Group

Lecture: Monday 3-4 CHI WAH Building(智华楼) 208

王中秋 (WANG Zhongqiu) wangzq3@sustech.edu.cn

Lab Course:

Group 1: Monday 9-10 CHI WAH Building(智华楼) 511

王中秋 (WANG Zhongqiu) wangzq3@sustech.edu.cn

TA: 宋一鸣 (SONG Yiming) 12432696@mail.sustech.edu.cn;

Group 2: Monday 7-8 CHI WAH Building(智华楼) 509

Lab Tutor: 朱悦铭 (ZHU Yueming) zhuym@sustech.edu.cn (Office Hour: Wednesday 15:00-17:00 in 工学院南楼111)

TA: 段晟俊 (DUAN Shengjun) 12632582@mail.sustech.edu.cn

Group 2: Wednesday 3-4 CHI WAH Building(智华楼) 509

Lab Tutor: 朱悦铭 (ZHU Yueming) zhuym@sustech.edu.cn (Office Hour: Wednesday 15:00-17:00 in 工学院南楼111)

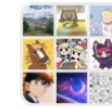
TA for second project:

TA: 黄亮 12542021@mail.sustech.edu.cn

Course Group: Wechat Enterprise

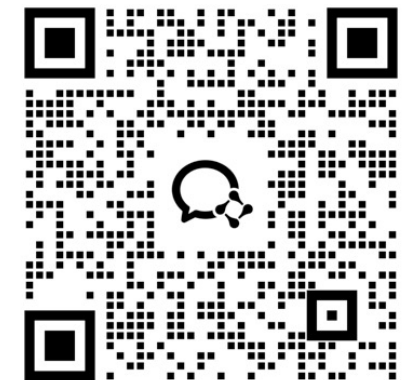
Lecture Notes, Blackboard and WeCom Group

- QR code for the WeCom group
 - Teaching assistants (TA), Yueming, and I will be there to help



CS307DB-course-group

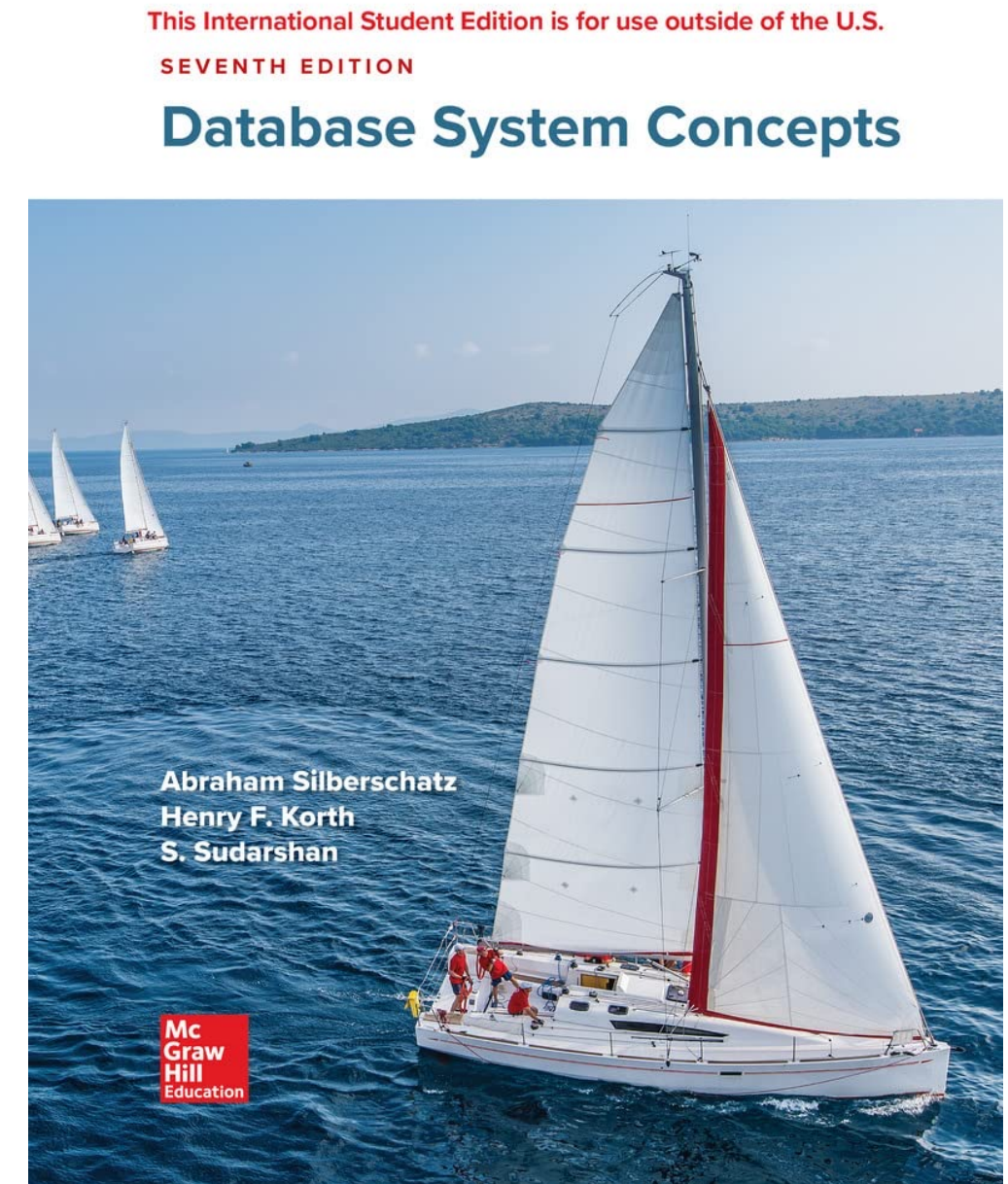
This is an internal group chat of the company. Only the company members can join it by scanning QR code



The QR code is valid before 09/14 and will be updated if you enter again.

Textbooks

- Reference book:
 - A. Silberschatz, H. Korth, and S. Sudarshan. **Database System Concepts**. McGraw-Hill, New York, 7th Edition, (2019).



Grading Policy

- Lecture and Lab Attendance (10%)
 - 5% for each part
- Assignments (15%)
 - 3 – 5 assignments
- Projects (40%)
 - 2 projects (approximately week 5 - 9, and week 10 - 14)
- Final exam (35%)

- 2026 Spring – statistics of final letter grades
 - Top 28% students scored A-/A/A+
 - Top 82% students scored at least B-/B/B+
 - Top 93% students scored at least C-/C/C+

Grading Policy

- Late Submission
 - **We do not accept late submissions**
 - All assignments, quizzes, and projects, etc. will receive **a score of zero if you miss the deadline**
 - Do not make amateur mistakes
- Groups for Projects
 - **Groups across different lab sessions are not allowed.** Find your teammate in the same lab session.
 - Find a teammate who is **responsible**
 - Unless you feel okay to work with someone who is always coasting, but gets the same grade as you
- Grading
 - The teachers and TAs guarantee that your assignments and projects will be evaluated carefully and unbiasedly
 - We do not accept arguing with teachers over a certain grade once the decision has been made

Plagiarism

- Please read the regulations on academic misconduct on the Blackboard site
 - Blackboard – “[Declaration Form Submission](#)”
 - All documents are in the attachment of the assignment
- Upload the signed *[Undergraduate Students Declaration Form](#)* to Blackboard with the following file name format:
 - SID_name.pdf
 - * The submission entry will be announced later on Blackboard
- **Do NOT plagiarize / copy-paste others' homework, code etc.**
 - Severe consequences
 - Be careful if using Generative AI

Some Other Stuff

- Computing technologies advance very fast
 - Search online to learn more by yourself
 - Search engines (Google, Bing, Baidu, etc.), StackOverflow, GitHub
 - Ask AI (要善于用 AI)
 - The lecture notes can guide your self study
- You are encouraged to ask questions
- Practice makes perfect

How much effort you should spend?

- My experience
 - I ranked 7th among 200 students in CS@HIT, according to 保研成绩
 - Before I started to teach database in FALL 2024, I forgot everything I learned about database in 2010
 - I spent ~1.5 days per week to fully understand each lecture
 - including course slides, and reading corresponding chapters in the textbook
 - Likely less effort, as the class is changed to elective

Principles of Database Systems (CS307)

Lecture 1: Introduction to Databases

Zhong-Qiu Wang

Department of Computer Science and Engineering
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What is a Database?

- Well, first, let's take a step back: What is data?

data noun, plural in form but singular or plural in construction, often attributive

 Save Word

da·ta | \ 'dā-tə , 'da-  also 'dä-  \

Essential Meaning of *data*

- 1** : facts or information used usually to calculate, analyze, or plan something
// She spent hours reviewing the *data* from the experiment.
// They made their decisions based on the survey *data*.
[See More Examples](#)
- 2** : information that is produced or stored by a computer
// She works as a *data* entry clerk.
// There was too much *data* for the computer to process.
// He is an expert in *data* retrieval. [=finding information stored on a computer]

What is a Database?

- A modern database system is a complex software system whose task is to manage a large, complex collection of data
 - Maintains a collection of interrelated data
 - E.g., in a class registration system, Student, Instructors, Classes, Classroom etc.
 - Provides a set of programs to access and modify the data
 - E.g., check how many female students select CS307
 - Change the classroom of CS307 to another place
 - Query who select what classes
 - Rank the students in each department according to their GPA
 - Provides an environment that is both convenient and efficient to use
 - Provide a language to support query, add, deletion, and modification
 - E.g., `select count(*) from student where student.gender = 'male'`

Example tables in relational database for class registration

Student ID	Student name	Gender	Born Year	Course ID	Instructor ID	Course name	Instructor Name	salary
0	张三	Male	1991	2	2	Operating system	张七	1100
1	李四	Female	1992	3	3	Algorithms	李五	3000
2	王麻子	Male	1991	1	1	Principals of database	曹六	1000
3	何二	Male	1993					
4	李一	Female	1995	3	3	Algorithms	李五	3000
4	李一	Female	1995	1	1	Principals of database	曹六	1000
4	李一	Female	1995	3	3	Algorithms	李五	3000
					4		王一	10000

Using large table is a good idea? Any problems?

Example tables in relational database for class registration

Student ID	Student name	Gender	Born Year	Course ID	Instructor ID	Course name	Instructor Name	salary
0	张三	Male	1991	2	2	Operating system	张七	1100
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4	李一	Female	1995	3	3	Algorithms	李五	3000
					4		王一	10000

1. **Data redundancy** (same values stored multiple times); 2. **Missing values** waste space

Example tables in relational database for class registration

Student ID	Student name	Gender	Born Year
0	张三	Male	1991
1	李四	Female	1992
2	王麻子	Male	1991
3	何二	Male	1993
4	李一	Female	1995

Student Table

Student ID	Course ID
0	2
1	3
2	1
4	3
4	1
4	2

Course_Registration

Course ID	Instructor ID	Course name
1	1	Principals of database
2	2	Operating system
3	3	Algorithms

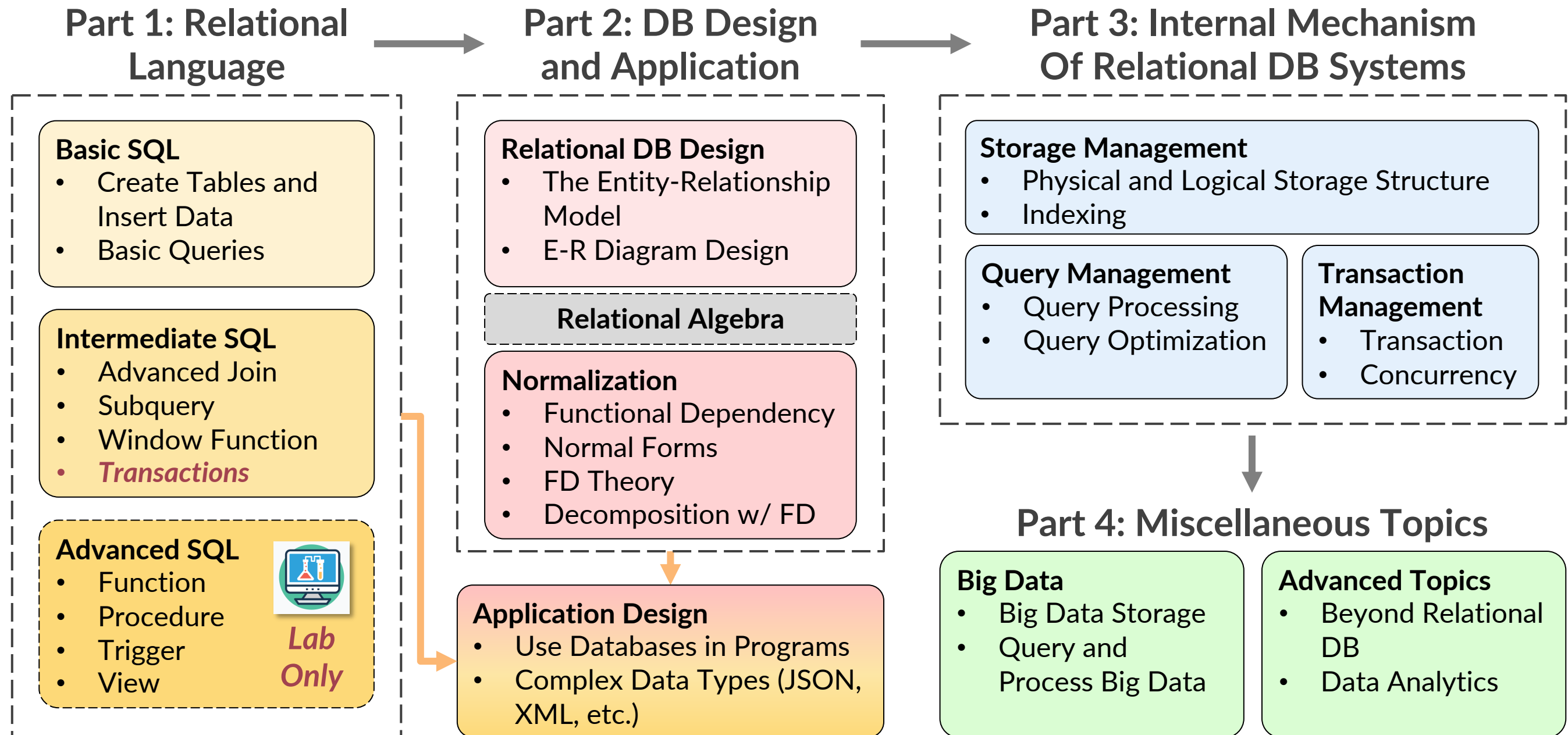
Course Table

Instructor ID	Instructor Name	salary
1	曹六	1000
2	张七	1100
3	李五	3000
4	王一	10000

Instructor Table

- Decompose into smaller tables a good idea?
 - How and whether to decompose?
 - Equivalent to original large table?
 - How to design the tables?
 - How to query the tables to get information we want?
 - How to store the tables efficiently on hard drives?
 - How to optimize the queries to reduce query cost?

Outline



Applications of Database

- Databases touch all aspects of our lives
- Enterprise Information
 - **Sales**: customers, products, purchases
 - **Accounting**: payments, receipts, assets
 - **Human Resources**: Information about employees, salaries, payroll taxes.
 - **Bank**: Information about employees, customers, transactions, bank branches
- Manufacturing
 - Management of production, inventory, orders, supply chain.
- Universities
 - Registration, grades



Applications of Database

- Databases are everywhere today
 - ... but the concept is old
 - The idea was to have one system doing once and for all the boring **data storage/retrieval** part
 - Hide how the data are stored and maintained
 - Like a warehouse, hidden details about how you store items, and provide a simple and easy-to-use interface for users



Purpose of Database Systems

- In the early days ...
 - Database applications were built directly on top of file systems
 - Operating-system files
 - + some applications processing the files
 - Add new students, instructors, and courses
 - Register students for courses and generate class rosters
 - Assign grades to students, compute grade point averages (GPA), and generate transcripts
 - When creating a new department, need to write code to update information and rules

Purpose of Database Systems

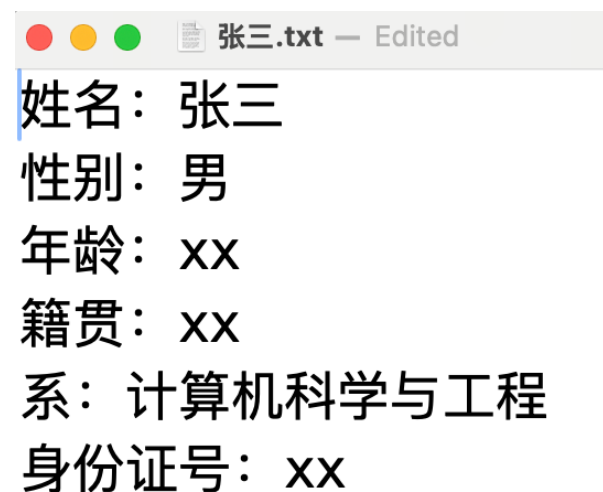
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Purpose of Database Systems

- In the early days ...
 - Database applications were built directly on top of file systems
 - Operating-system files + some applications processing the files
 - However, it suffers from many issues, including (but not limited to):
 - Data redundancy and inconsistency
 - Difficulty in accessing data
 - Data isolation
 - Integrity problems
 - Atomicity of updates
 - Concurrent access by multiple users
 - Security problems

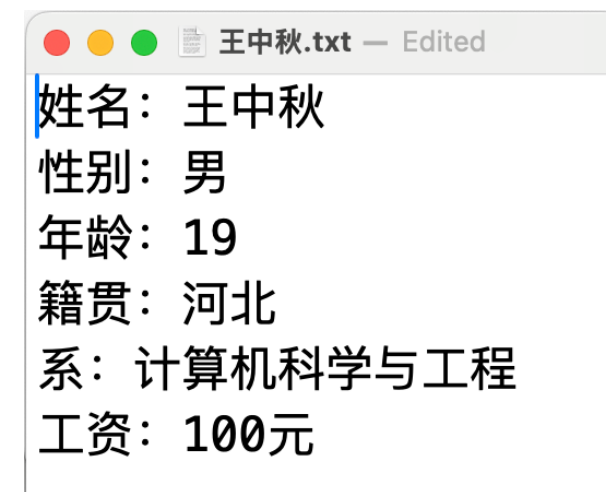
Purpose of Database Systems

- In the early days ...
 - Database applications were built directly on top of file systems
 - Operating-system files + some applications processing the files
 - However, it suffers from many issues, including (but not limited to):
 - Data redundancy and inconsistency
 - Data is stored in multiple file formats, resulting in duplication of information in different files
 - Only changing a subset of the files leads to inconsistency



张三.txt — Edited

姓名：张三
性别：男
年龄：xx
籍贯：xx
系：计算机科学与工程
身份证号：xx



王中秋.txt — Edited

姓名：王中秋
性别：男
年龄：19
籍贯：河北
系：计算机科学与工程
工资：100元

Purpose of Database Systems

- In the early days ...
 - Database applications were built directly on top of file systems
 - Operating-system files + some applications processing the files
 - However, it suffers from many issues, including (but not limited to):
 - Difficulty in accessing data
 - Need to write a new program to carry out each new task
 - » E.g., find the average salary of the female instructors in the computer science department who is older than 40
 - Conventional systems do not allow needed data to be retrieved in a convenient and efficient manner

Purpose of Database Systems

- In the early days ...
 - Database applications were built directly on top of file systems
 - Operating-system files + some applications processing the files
 - However, it suffers from many issues, including (but not limited to):
 - **Data isolation**
 - Data are scattered in various files
 - Files may be in different formats
 - Writing new application programs to retrieve the appropriate data is difficult

Purpose of Database Systems

- In the early days ...
 - Database applications were built directly on top of file systems
 - Operating-system files + some applications processing the files
 - However, it suffers from many issues, including (but not limited to):
 - Integrity problems
 - Integrity constraints (e.g., salary > 0) become “buried” in program code rather than being stated explicitly
 - Hard to add new constraints or change existing ones
 - Problem is compounded when constraints involve several data items from different files.



王中秋.txt — Edited

姓名：王中秋
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工资：100元

Purpose of Database Systems

- In the early days ...
 - Database applications were built directly on top of file systems
 - Operating-system files + some applications processing the files
 - However, it suffers from many issues, including (but not limited to):
 - Atomicity of updates
 - Failures could happen anytime
 - Failures may leave database in an inconsistent state with partial updates carried out
 - » E.g., transfer of funds from one account to another should either complete or not happen at all

Purpose of Database Systems

- In the early days ...
 - Database applications were built directly on top of file systems
 - Operating-system files + some applications processing the files
 - However, it suffers from many issues, including (but not limited to):
 - Concurrent access by multiple users
 - Concurrent access needed for performance (e.g., banking systems)
 - Uncontrolled concurrent accesses can lead to inconsistencies
 - » Example: Two people reading a balance (say 100) at the same time and updating it by withdrawing money (say 50 each) at the same time
 - » Remaining balance should be 0 rather than 50

Purpose of Database Systems

- In the early days ...
 - Database applications were built directly on top of file systems
 - Operating-system files + some applications processing the files
 - However, it suffers from many issues, including (but not limited to):
 - Security problems
 - Not every user of the database system should be able to access all the data
 - » Example: payroll personnel can only see financial information. No access to information about academic records
 - Enforcing such security constraints is difficult
 - Read/insert/update/delete authorization

Purpose of Database Systems

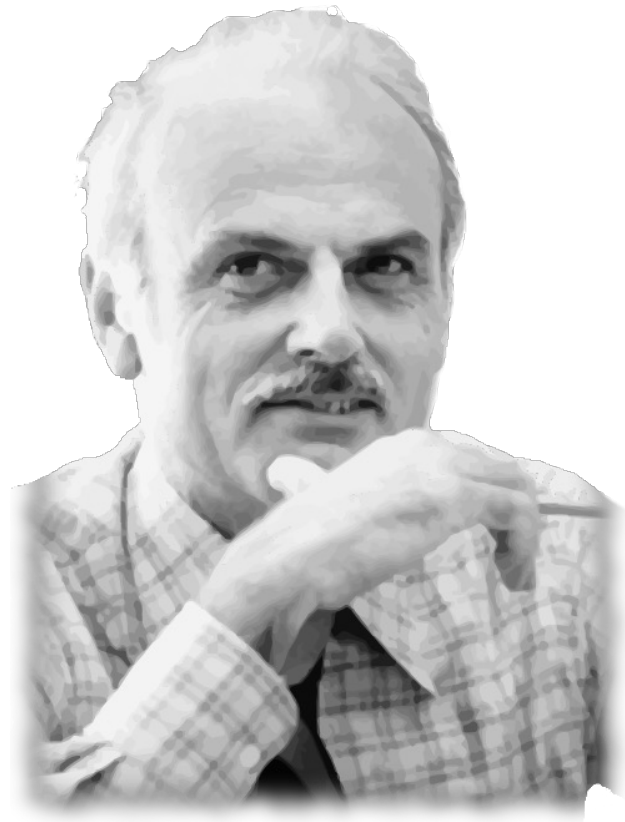
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 - However, it suffers from many issues, including (but not limited to):
 - Data redundancy and inconsistency
 - Difficulty in accessing data
 - Data isolation
 - Integrity problems
 - Atomicity of updates
 - Concurrent access by multiple users
 - Security problems

Database systems offer solutions to all the problems mentioned above

A Bit of History

- 1950s and early 1960s:
 - Data processing using magnetic tapes for storage
 - Tapes provided only sequential access
 - Punched cards for input
- Late 1960s and 1970s:
 - Hard disks allowed direct access to data
 - Network and hierarchical data models in widespread use
 - **Ted Codd** defines the **relational data model**
 - Won the ACM Turing Award for this work
 - IBM Research begins System R prototype
 - UC Berkeley (Michael Stonebraker) begins Ingres prototype
 - **Oracle** releases first commercial relational database
 - High-performance (for the era) transaction processing

A Bit of History



Edgar F. “Ted” Codd
1923 – 2003

Turing Award 1981

A Relational Model of Data for Large Shared Data Banks

E. F. CODD

IBM Research Laboratory, San Jose, California

Future users of large data banks must be protected from having to know how the data is organized in the machine (the internal representation). A prompting service which supplies such information is not a satisfactory solution. Activities of users at terminals and most application programs should remain unaffected when the internal representation of data is changed and even when some aspects of the external representation are changed. Changes in data representation will often be needed as a result of changes in query, update, and report traffic and natural growth in the types of stored information.

Existing noninferential, formatted data systems provide users with tree-structured files or slightly more general network models of the data. In Section 1, inadequacies of these models are discussed. A model based on n -ary relations, a normal form for data base relations, and the concept of a universal

The relational view (or model) of data described in Section 1 appears to be superior in several respects to the graph or network model [3, 4] presently in vogue for non-inferential systems. It provides a means of describing data with its natural structure only—that is, without superimposing any additional structure for machine representation purposes. Accordingly, it provides a basis for a high level data language which will yield maximal independence between programs on the one hand and machine representation and organization of data on the other.

A further advantage of the relational view is that it forms a sound basis for treating derivability, redundancy, and consistency of relations—these are discussed in Section 2. The network model, on the other hand, has spawned a number of confusions, not the least of which is mistaking the derivation of connections for the derivation of relations (see remarks in Section 2 on the “connection trap”).

Finally, the relational view permits a clearer evaluation of the scope and logical limitations of present formatted data systems, and also the relative merits (from a logical standpoint) of competing representations of data within a single system. Examples of this clearer perspective are cited in various parts of this paper. Implementations of systems to support the relational model are not discussed.

E. F. Codd, A Relational Model of Data for Large Shared Data Banks, *Information Retrieval*, June, 1970

A Bit of History

- 1980s:
 - Research relational prototypes evolve into commercial systems
 - SQL becomes industrial standard
 - Parallel and distributed database systems
 - Wisconsin, IBM, Teradata
 - Object-oriented database systems
- 1990s:
 - Large decision support and data-mining applications
 - Large multi-terabyte data warehouses
 - Emergence of Web commerce

A Bit of History

- 2000s
 - Big data storage systems
 - Google BigTable, Yahoo PNuts, Amazon,
 - “NoSQL” systems.
 - Big data analysis: beyond SQL
 - Map reduce and friends
- 2010s
 - SQL reloaded
 - SQL front end to Map Reduce systems
 - Massively parallel database systems
 - Multi-core in-memory databases

Relational Database

- Based on the relational model of data
 - Organizes data into one or more tables
 - Rows are also called records or tuples
 - Columns are also called attributes

Columns (Attributes)

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

Rows (Tuples)

(a) The *instructor* table

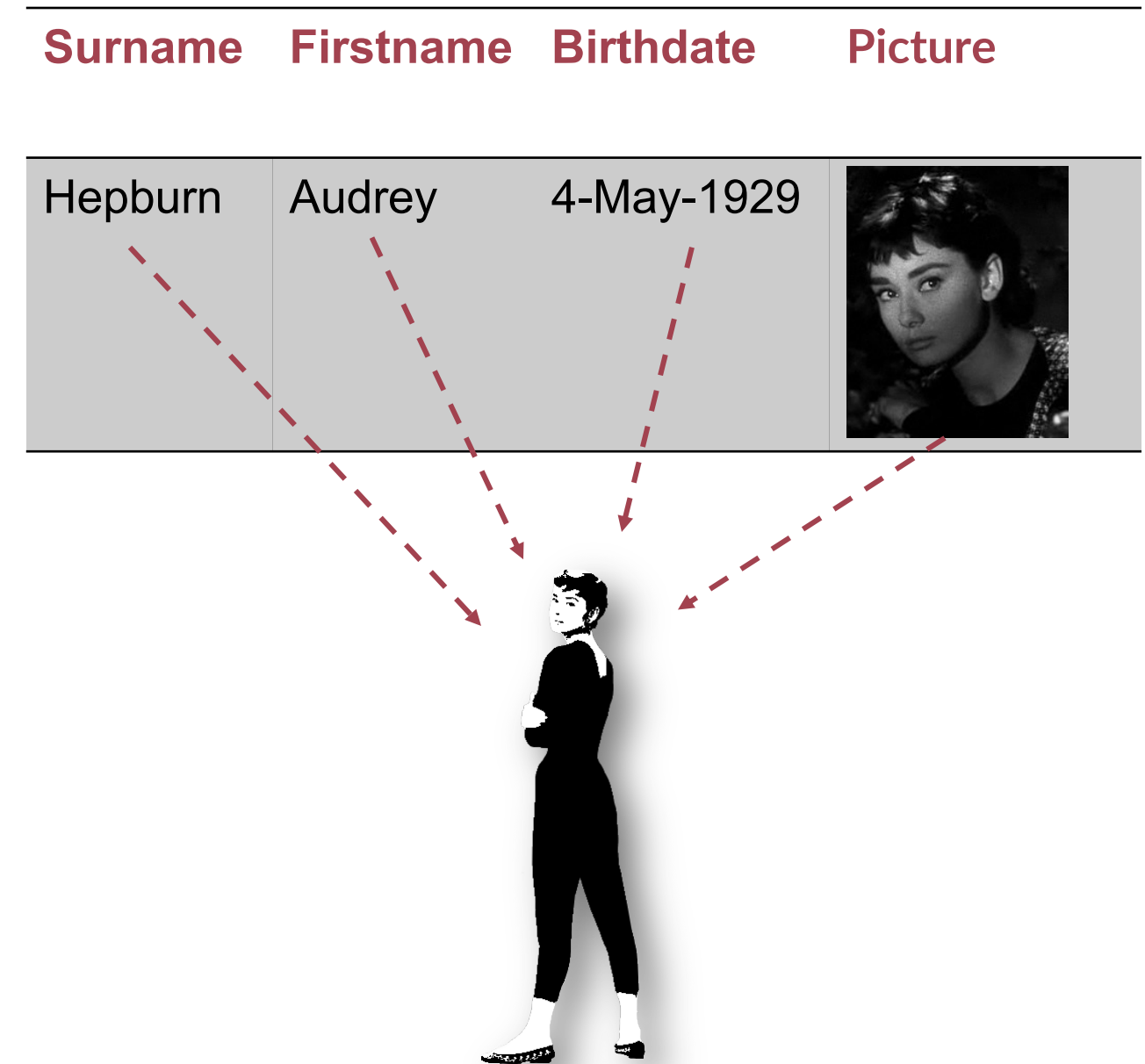
Relational Database

- Each column stores a piece of data
- One row represents a “known fact”:
 - “Audrey Hepburn was born on 1929/05/04 and looked like this.”

Surname	Firstname	Birthdate	Picture
Hepburn	Audrey	4-May-1929	

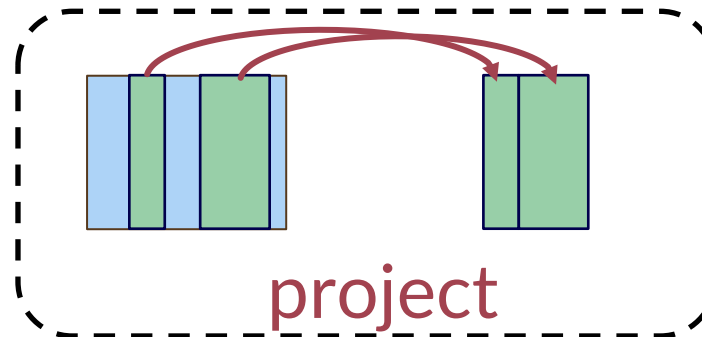
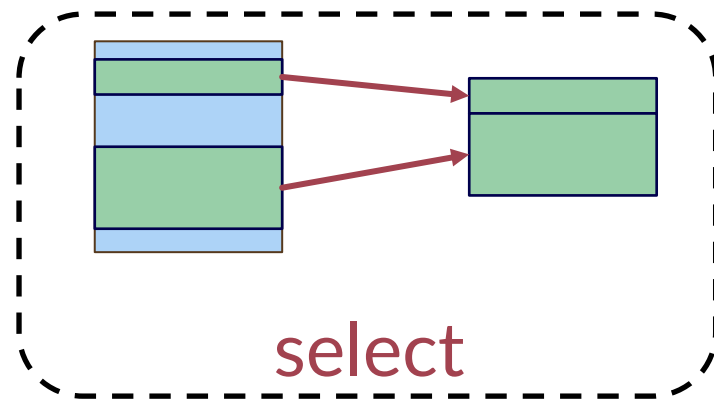
Relational Database

- Each column stores a piece of data
- One row represents a “known fact”:
 - “Audrey Hepburn was born on 1929/05/04 and looked like this.”
- All the pieces of data in a row are related, hence "**relational**"



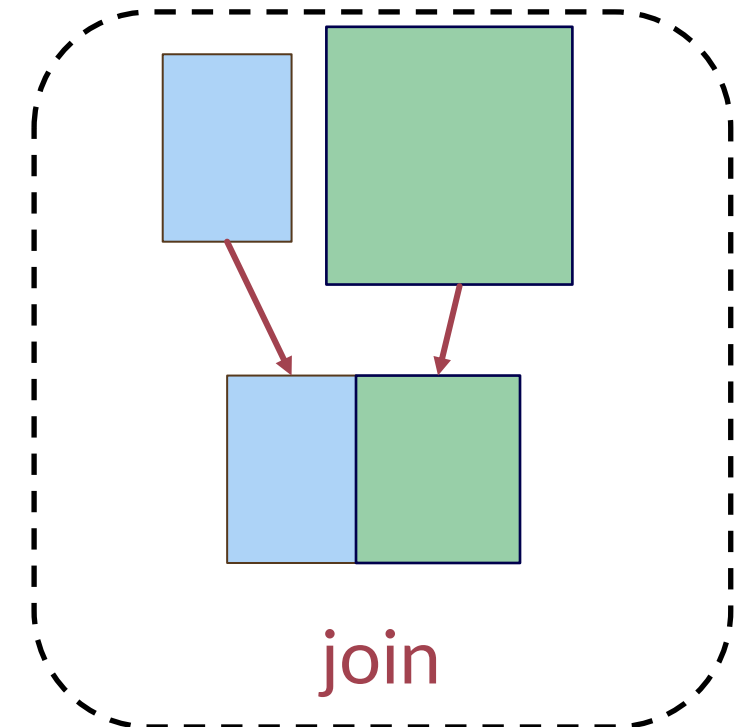
Relational Database

- But Codd's "big idea"
 - You could operate on the relations and get new sets
- Relational Algebra
 - Theoretical foundation for relational databases



Student(Student_ID, gender)

StudentClass(Student_ID, Class_ID)



E.g., Find #male students in each class

Key (键)

- Example: A Film Database
 - Easy to find such as “The 100 greatest films ever”
 - Sometimes as a .csv file that you can load into a spreadsheet

1	Movie Title	Country	Year	Director	Starring	
2	Citizen Kane	US	1941	welles, o.	Orson Welles, Joseph Cotten	
3	La règle du jeu	FR	1939	Renoir, J.	Roland Toutain, Nora Grégor, Marcel Dalio, Jean Renoir	
4	North by Northwest	US	1959	HITCHCOCK, A.	Cary GRANT, Eva Marie SAINT, James MASON	
5	Singin' In the Rain	US	1952	Donen/Kelly	Gene Kelly, Debbie Reynolds, Donald O'Connor	
6	Rear Window	US	1954	HITCHCOCK, A.	James STEWART, Grace KELLY	
7	City Lights	US	1931	CHAPLIN, C.	Charlie CHAPLIN, Virginia CHERRILL	
8	The Third Man	GB	1949	Reed, C.	Joseph Cotten, Alida Valli, Orson Welles	
9	The Searchers	US	1956	Ford, J.	John Wayne, Jeffrey Hunter, Natalie Wood	
10	Ladri di biciclette	IT	1949	DeSica, V.	Lamberto Maggiorani, Enzo Staiola	
11	Annie Hall	US	1977	Allen, W.	Woody Allen, Diane Keaton	
12	On the Waterfront	US	1954	Kazan, E.	Marlon Brando, Eva Marie Saint, Karl Malden	
13	All about Eve	US	1950	Mankiewicz, J.	Bette Davis, Anne Baxter, George Sanders	
14	Casablanca	US	1942	Curtiz, M.	Humphrey Bogart, Ingrid Bergman, Claude Rains	
15	The Treasure of the Sierra Madre	US	1948	HUSTON, J.	Humphrey BOGART, Walter HUSTON, Tim HOLT	
16	High Noon	US	1952	Zinnemann, F.	Gary Cooper, Grace Kelly	
17	Some Like It Hot	US	1959	Wilder, B.	Tony Curtis, Jack Lemmon, Marilyn Monroe	
18						

Key

- Duplicates are forbidden in relational tables
 - Introduces potential errors such as when counting the number of movies

Movie Title	Country	Year	Director	Starring
Citizen Kane	US	1941	welles, o.	Orson Welles, Joseph Cotten
La règle du jeu	FR	1939	Renoir, J.	Roland Toutain, Nora Grégor, Marcel Dalio, Jean Renoir
North By Northwest	US	1959	HITCHCOCK, A.	Cary Grant, Eva Marie Saint, James Mason
Singin' in the Rain	US	1952	Donen/Kelly	Gene Kelly, Debbie Reynolds, Donald O'Connor
Rear Window	US	1954	HITCHCOCK, A.	James Stewart, Grace Kelly
North By Northwest	US	1959	HITCHCOCK, A.	Cary Grant, Eva Marie Saint, James Mason

Key

- How to identify “different rows” in a table?
 - A column (or a set of columns) to **differentiate** one row from another
 - In the film data ... How about the movie titles?

Movie Title	Country	Year	Director	Starring
Citizen Kane	US	1941	welles, o.	Orson Welles, Joseph Cotten
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Singin' in the Rain	US	1952	Donen/Kelly	Gene Kelly, Debbie Reynolds, Donald O'Connor
Rear Window	US	1954	HITCHCOCK, A.	James Stewart, Grace Kelly

搜索 神雕侠侣



神雕侠侣 神雕侠侣 (1995) [剧集] [可播放]

★★★★★ 9.2 (176564人评价)

中国香港 / 爱情 / 武侠 / 古装 / 新神雕侠侣 / Return Of The Condor Heroes / 45分钟
李添胜 / 古天乐 / 李若彤 / 白彪 / 魏秋桦 / 傅明宪 / 李绮虹 / 雪梨 / 简佩筠

话题 《神雕侠侣》哪个角色最让你惊喜？
10030人浏览 · 22篇文章



神雕侠侣 (2006) [剧集] [可播放]

★★★★★ 7.5 (70883人评价)

中国大陆 / 武侠 / 古装 / The Return of the Condor Heroes / 40分钟
于敏 / 刘亦菲 / 黄晓明 / 陈紫函 / 杨幂 / 叮当 / 王洛勇 / 赵鸿飞 / 钱博



神雕侠侣 (2014) [剧集] [可播放]

★★★★★ 4.9 (26245人评价)

中国大陆 / 剧情 / 武侠 / 新神雕侠侣 / The Condor Heroes / 45分钟
李慧珠 / 邓伟恩 / 李达超 / 陈晓 / 陈妍希 / 张馨予 / 张雪迎 / 郑国霖 / 杨明娜 / 陈翔 / 毛晓彤



新神雕侠侣 (2022) [剧集]

★★★★★ (尚未播出)

中国大陆 / 爱情 / 武侠 / 古装
林峰 / 佟梦实 / 毛晓慧 / 文淇 / 涂冰 / 邵兵 / 龚蓓苾 / 毛林林 / 宗峰岩



神雕侠侣 (2023)

★★★★★ (尚未上映)

中国大陆 / 剧情 / 爱情 / 武侠 / 古装 / The Romance Of The Condor Heros
徐克



神雕侠侣 神雕侠侣 (1983) [剧集] [可播放]

★★★★★ 7.9 (6328人评价)

中国香港 / 剧情 / 动作 / 爱情 / The Return of the Condor Heroes / 42分钟
范秀明 / 鞠觉亮 / 萧显辉 / 司徒立光 / 谭锐铭 / 刘德华 / 陈玉莲 / 梁家仁 / 欧阳佩珊 / 廖安丽 / 吕有慧 / 曾江



神雕侠侣 (1998) [剧集] [可播放]

★★★★★ 7.1 (5767人评价)

新加坡 / 爱情 / 武侠 / 古装 / 神雕侠侣 98版 / Return of the Condor Heroes / 47分钟
马玉辉 / 谢敏洋 / 蔡晶盛 / 张龙敏 / 卢燕金 / 李铭顺 / 范文芳 / 朱厚任 / 何咏芳 / 林湘萍 / 丁岚 / 李南星



神雕侠侣 (1998) [剧集] [可播放]

★★★★★ 5.2 (10379人评价)

中国台湾 / 中国大陆 / 武侠 / 古装 / 杨过与小龙女 / 45分钟
李惠民 / 赖水清 / 任贤齐 / 吴倩莲 / 孙兴 / 季芹 / 李立群 / 夏文汐 / 蔡君茹 / 高捷



神雕侠侣 (2001) [剧集] [可播放]

★★★★★ 7.5 (1772人评价)

日本 / 中国香港 / 动画 / 24分钟
案纳正美 / 高木淳 / 浪川大辅 / 园崎未惠 / 中田让治 / 唐泽润 / 木村亚希子 / 小村哲生 / 广田行生 / 高户靖广



神雕侠侣 (1984) [剧集]

★★★★★ 7.5 (926人评价)

中国台湾 / 剧情 / 爱情 / 武侠 / 古装 / 95分钟
何东兴 / 孟飞 / 潘迎紫 / 傅娟



九一神雕侠侣 九一神鵰侠侣 (1991) [可播放]

★★★★★ 6.4 (8789人评价)

中国香港 / 动作 / 剧情 / 奇幻 / 爱情 / 科幻 / 神秘英豪 / 新神雕侠侣 / 92分钟
元奎 / 黎大炜 / 刘德华 / 梅艳芳 / 郭富城 / 叶蕴仪 / 刘嘉玲



神雕侠侣 神鵰侠侣 (1982) [可播放]

★★★★★ 5.2 (1402人评价)

中国香港 / 动作 / 爱情 / 武侠 / 古装 / 射雕英雄传4 / Brave Archer 4 / 100分钟
张彻 / 江生 / 郭追 / 傅声 / 龙天翔 / 黄淑仪



九二神雕之痴心情长剑 九二神鵰之痴心情長劍 (1992) [可播放]

★★★★★ 6.0 (6152人评价)

中国香港 / 喜剧 / 爱情 / 奇幻 / 武侠 / 神秘情侠 / 新神雕侠侣2(台) / 92分钟
元奎 / 黎大炜 / 刘德华 / 关之琳 / 吴耀汉 / 关淑怡



神雕侠侣 (1976) [剧集]

★★★★★ (暂无评分)

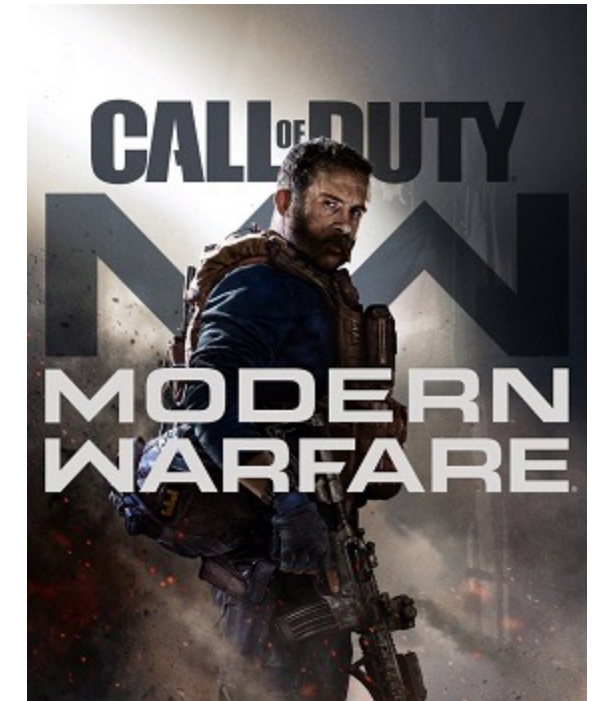
中国香港 / 古装
萧笙 / 罗乐林 / 李通明 / 白彪 / 米雪 / 曾江 Kenneth Tsang / 秦煌 / 郑裕玲 / 冯淬帆

Key

- How to identify “different rows” in a table?
 - A column (or a set of columns) to **differentiate** one row from another
 - In the film data ... ~~How about the movie titles?~~ Title + Director?

Movie Title	Country	Year	Director	Starring
Citizen Kane	US	1941	welles, o.	Orson Welles, Joseph Cotten
La règle du jeu	FR	1939	Renoir, J.	Roland Toutain, Nora Grégor, Marcel Dalio, Jean Renoir
North By Northwest	US	1959	HITCHCOCK, A.	Cary Grant, Eva Marie Saint, James Mason
Singin' in the Rain	US	1952	Donen/Kelly	Gene Kelly, Debbie Reynolds, Donald O'Connor
Rear Window	US	1954	HITCHCOCK, A.	James Stewart, Grace Kelly

But, Hitchcock did It (So it was with the COD game series)



Same developer (Infinite Ward)
Same publisher (Activision)
Two different years (2007, 2019)

Key

- How to identify “different rows” in a table?
 - A column (or a set of columns) to **differentiate** one row from another
 - In the film data ... ~~How about the movie titles? Title + Director?~~
 - Title + Director + Year?

Movie Title	Country	Year	Director	Starring
Citizen Kane	US	1941	welles, o.	Orson Welles, Joseph Cotten
La règle du jeu	FR	1939	Renoir, J.	Roland Toutain, Nora Grégor, Marcel Dalio, Jean Renoir
North By Northwest	US	1959	HITCHCOCK, A.	Cary Grant, Eva Marie Saint, James Mason
Singin' in the Rain	US	1952	Donen/Kelly	Gene Kelly, Debbie Reynolds, Donald O'Connor
Rear Window	US	1954	HITCHCOCK, A.	James Stewart, Grace Kelly

Key

- Ok, you have made it this time, but –
 - The combination is too difficult for either remembering or computing
 - Need to compare multiple times on different columns
 - Think about deduplication
 - What if there are multiple items in a single column?
 - For example, more than one directors in a movie?



Primary Key (主键)

Additional material about creating a unique ID for each row:
https://en.wikipedia.org/wiki/Universally_unique_identifier

- Some of the keys may be unique for every row
 - Student ID, Email address, 18-digit ID number, etc.
- Usually, it is a good practice to choose the simplest one
 - (Or, create one)

Movie ID	Movie Title	Country	Year	Director	Starring
0	Citizen Kane	US	1941	welles, o.	Orson Welles, Joseph Cotten
1	La règle du jeu	FR	1939	Renoir, J.	Roland Toutain, Nora Grégor, Marcel Dalio, Jean Renoir
2	North By Northwest	US	1959	HITCHCOCK, A.	Cary Grant, Eva Marie Saint, James Mason
3	Singin' in the Rain	US	1952	Donen/Kelly	Gene Kelly, Debbie Reynolds, Donald O'Connor
4	Rear Window	US	1954	HITCHCOCK, A.	James Stewart, Grace Kelly

Normalization

- A way of standardize your data

Movie ID	Movie Title	Country	Year	Director	Starring
0	Citizen Kane	US	1941	welles, o.	Orson Welles, Joseph Cotten
1	La règle du jeu	FR	1939	Renoir, J.	Roland Toutain, Nora Grégor, Marcel Dalio, Jean Renoir
2	North By Northwest	US	1959	HITCHCOCK, A.	Cary Grant, Eva Marie Saint, James Mason
3	Singin' in the Rain	US	1952	Donen/Kelly	Gene Kelly, Debbie Reynolds, Donald O'Connor
4	Rear Window	US	1954	Alfred Hitchcock	James Stewart, Grace Kelly

*Too many different
ways of spelling*

Normalization

- “First Norm Rule” (1NF, 第一范式)
 - Each column should only contain ONE piece of information

Director	Starring
welles, o.	Orson Welles, Joseph Cotten
Renoir, J.	Roland Toutain, Nora Grégor, Marcel Dalio, Jean Renoir
HITCHCOCK, A.	Cary Grant, Eva Marie Saint, James Mason
Donen/Kelly	Gene Kelly, Debbie Reynolds, Donald O'Connor
HITCHCOCK, A.	James Stewart, Grace Kelly



Director_Firstname	Director_Lastname
Alfred	Hitchcock
Orson	Welles

Starring_Firstname	Starring_Lastname
Orson	Welles
Joseph	Cotten

Normalization

- “First Norm Rule” (1NF, 第一范式)
 - Each column should only contain ONE piece of information

Director	Starring
welles, o.	Orson Welles, Joseph Cotten
Renoir, J.	Roland Toutain, Nora Grégor, Marcel Dalio, Jean Renoir
HITCHCOCK, A.	Cary Grant, Eva Marie Saint, James Mason
Donen/Kelly	Gene Kelly, Debbie Reynolds, Donald O'Connor
HITCHCOCK, A.	James Stewart, Grace Kelly



Director_ Firstname	Director_ Lastname	Born	Died
Alfred	Hitchcock	1899	1980
Orson	Welles	1915	1985

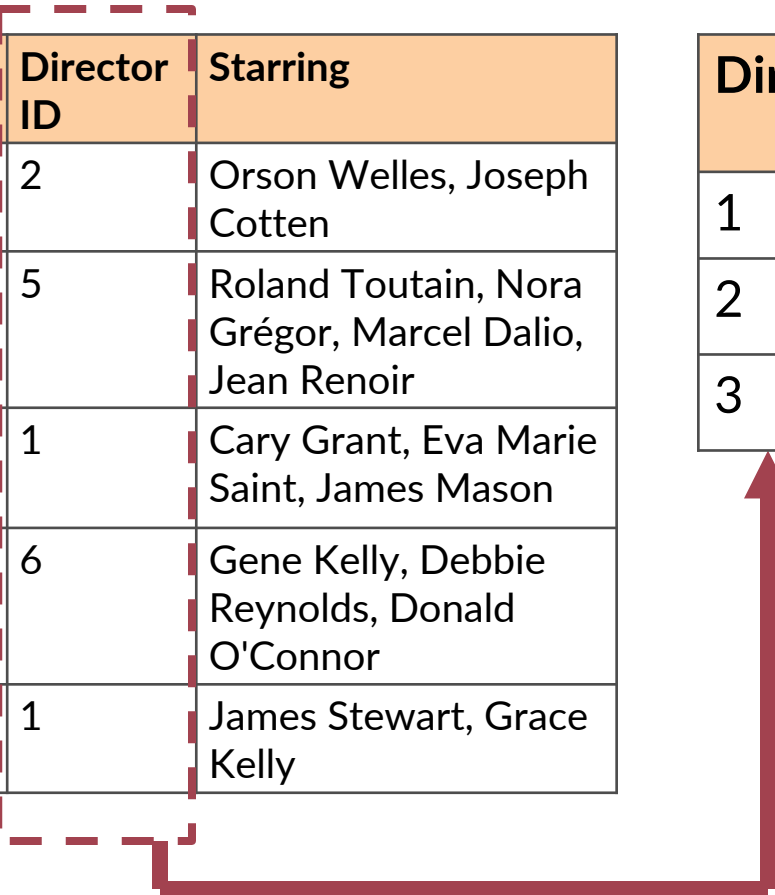
Extend the table to represent all directors

Normalization

- “First Norm Rule” (1NF, 第一范式)
 - Each column should only contain ONE piece of information

Movie ID	Movie Title	Country	Year	Director ID	Starring
0	Citizen Kane	US	1941	2	Orson Welles, Joseph Cotten
1	La règle du jeu	FR	1939	5	Roland Toutain, Nora Grégor, Marcel Dalio, Jean Renoir
2	North By Northwest	US	1959	1	Cary Grant, Eva Marie Saint, James Mason
3	Singin' in the Rain	US	1952	6	Gene Kelly, Debbie Reynolds, Donald O'Connor
4	Rear Window	US	1954	1	James Stewart, Grace Kelly

Director ID	Director_Firstname	Director_Lastname	Born	Died
1	Alfred	Hitchcock	1899	1980
2	Orson	Welles	1915	1985
3	...	...	...	...



Link the director information between tables

Normalization

- Normal Form (NF)
 - 1NF: Simple attributes
 - 2NF: Attributes depend on the full key
 - 3NF: Non-key attributes do not depend on each other
 - And many others

	UNF (1970)	1NF (1970)	2NF (1971)	3NF (1971)	EKNF (1982)	BCNF (1974)	4NF (1977)	ETNF (2012)	5NF (1979)	DKNF (1981)	6NF (2003)
Primary key (no duplicate tuples) ^[4]	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Atomic columns (cells cannot have tables as values) ^[5]	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Every non-trivial functional dependency either does not begin with a proper subset of a candidate key or ends with a prime attribute (no partial functional dependencies of non-prime attributes on candidate keys) ^[5]	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓	✓
Every non-trivial functional dependency either begins with a superkey or ends with a prime attribute (no transitive functional dependencies of non-prime attributes on candidate keys) ^[5]	✗	✗	✗	✓	✓	✓	✓	✓	✓	✓	✓
Every non-trivial functional dependency either begins with a superkey or ends with an elementary prime attribute	✗	✗	✗	✗	✓	✓	✓	✓	✓	✓	N/A
Every non-trivial functional dependency begins with a superkey	✗	✗	✗	✗	✗	✓	✓	✓	✓	✓	N/A
Every non-trivial multivalued dependency begins with a superkey	✗	✗	✗	✗	✗	✗	✓	✓	✓	✓	N/A
Every join dependency has a superkey component ^[8]	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓	N/A
Every join dependency has only superkey components	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	N/A
Every constraint is a consequence of domain constraints and key constraints	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗
Every join dependency is trivial	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓

Normalization

Question:
What if there are multiple director?



- “First Norm Rule” (1NF, 第一范式)
 - Each column should only contain ONE piece of information

Movie ID	Movie Title	Country	Year	Director ID	Starring
0	Citizen Kane	US	1941	2	Orson Welles, Joseph Cotten
1	La règle du jeu	FR	1939	5	Roland Toutain, Nora Grégor, Marcel Dalio, Jean Renoir
2	North By Northwest	US	1959	1	Cary Grant, Eva Marie Saint, James Mason
3	Singin' in the Rain	US	1952	6	Gene Kelly, Debbie Reynolds, Donald O'Connor
4	Rear Window	US	1954	1	James Stewart, Grace Kelly

Director ID	Director_Firstname	Director_Lastname	Born	Died
1	Alfred	Hitchcock	1899	1980
2	Orson	Welles	1915	1985
3	...	...	...	...

Link the director information
between tables

Entity and Relationship

- A bad idea: Add more columns in the Movie table

Movie ID	Movie Title	Country	Year	Director ID	Director 2 ID	Director 3 ID	Starring
----------	-------------	---------	------	-------------	---------------	---------------	----------

- Waste of space (not too many movies have 3 directors, let alone 6)

Entity and Relationship

- A bad idea: Add more columns in the Movie table

Movie ID	Movie Title	Country	Year	Director ID	Director 2 ID	Director 3 ID	Starring
----------	-------------	---------	------	-------------	---------------	---------------	----------

- Waste of space (not too many movies have 3 directors, let alone 6)
- How about starring? 10+ more columns?



Entity and Relationship

- Further refactoring of the tables ...

Movie ID	Movie Title	Countr y	Year
0	Citizen Kane	US	1941
1	La règle du jeu	FR	1939
2	North By Northwest	US	1959
3	Singin' in the Rain	US	1952
4	Rear Window	US	1954

Movie Entities

Relationship?

Director ID	Director_ Firstname	Director_ Lastname	Born	Died
1	Alfred	Hitchcock	1899	1980
2	Orson	Welles	1915	1985
3		...	...	...

Director Entities

Entity and Relationship

- Further refactoring of the tables ...

Movie ID	Movie Title	Countr y	Year
0	Citizen Kane	US	1941
1	La règle du jeu	FR	1939
2	North By Northwest	US	1959
3	Singin' in the Rain	US	1952
4	Rear Window	US	1954

Movie Entities

Directed By

Movie ID	Director ID
0	2
1	5
2	1

Relationship!

Director ID	Director_Firstname	Director_Lastname	Born	Died
1	Alfred	Hitchcock	1899	1980
2	Orson	Welles	1915	1985
3		...	...	...

Director Entities

Entity and Relationship

- Further refactoring of the tables ...

Movie ID	Movie Title	Country	Year
0	Citizen Kane	US	1941
1	La règle du jeu	FR	1939
2	North By Northwest	US	1959
3	Singin' in the Rain	US	1952
4	Rear Window	US	1954

Movie Entities

Directed By	
Movie ID	Director ID
0	2
1	5
2	1
...	...
16	8
16	9
16	10

Director ID	Director_Firstname	Director_Lastname	Born	Died
1	Alfred	Hitchcock	1899	1980
2	Orson	Welles	1915	1985
3		...	...	...

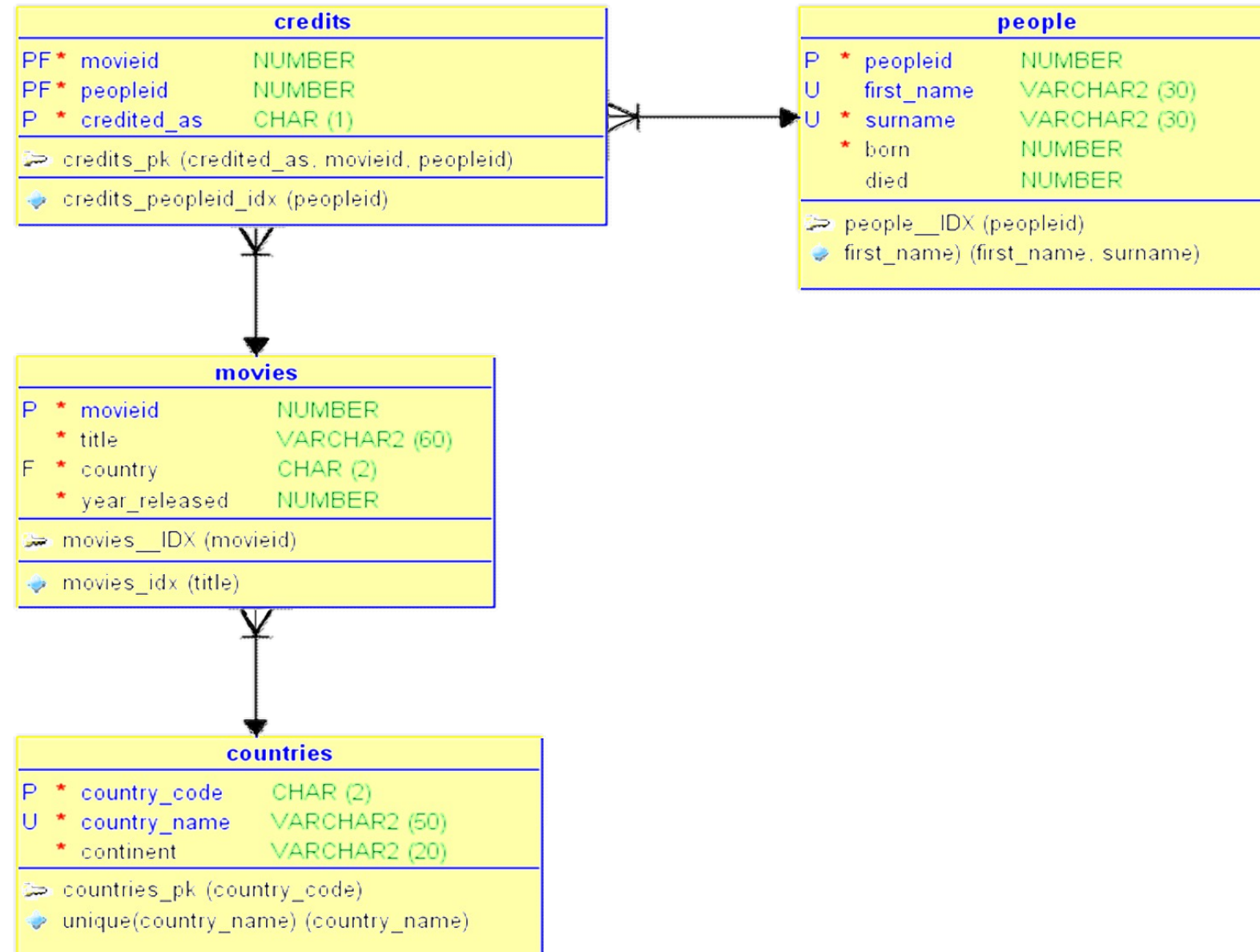
Director Entities

Question:
What if there are multiple director?

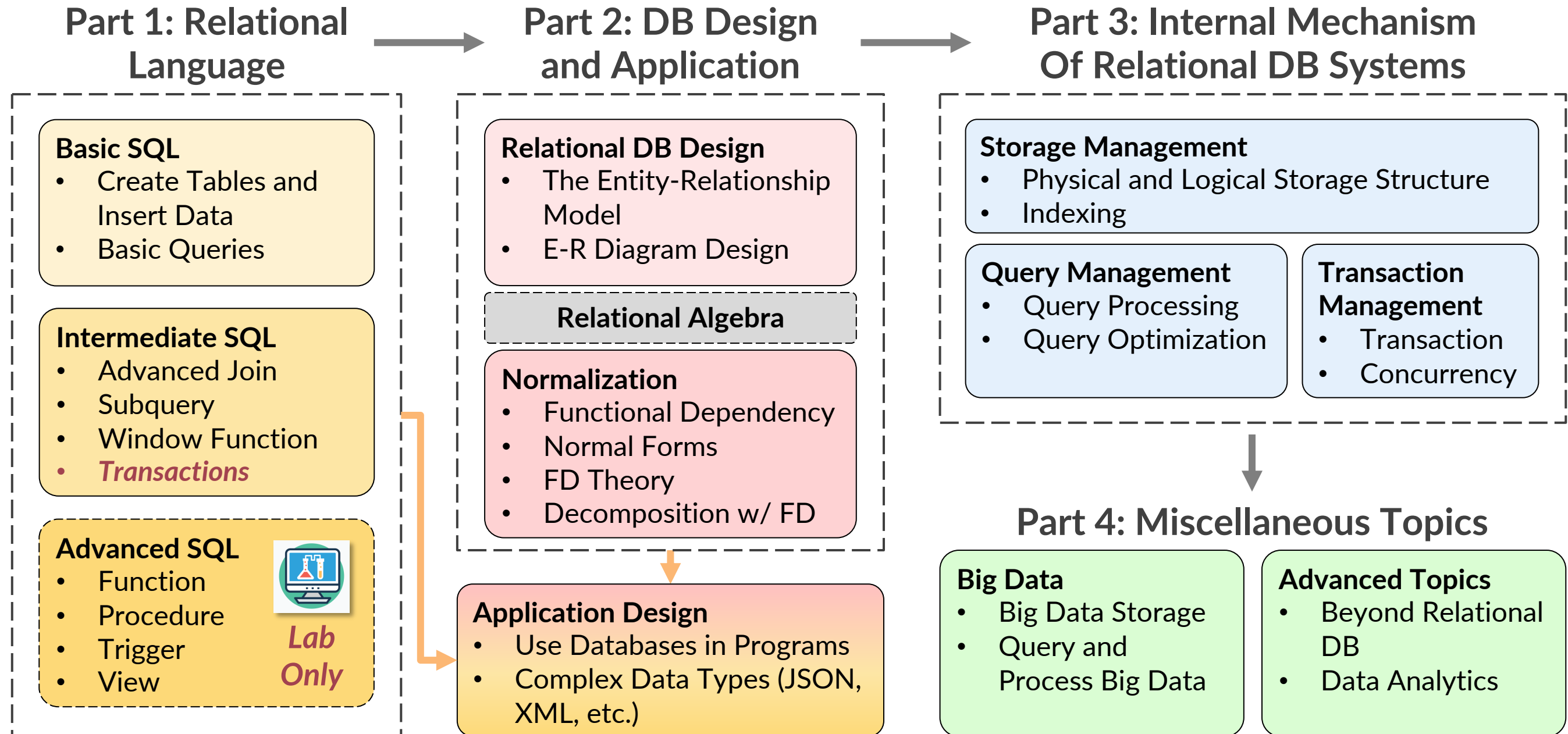
Answer:
Multiple rows in the relationship!

Entity and Relationship

- Starring -> Actor table
- Country -> Country table
 - You can also link the movies with corresponding actors, countries/regions, etc.
- Entity Relationship Diagram (E/R Diagram, ER Diagram, ERD)
 - A way of representing entity tables and their relationships (relationship tables)



Outline



Outline

- What we may not (fully) cover
 - Programming Language Support
 - We will focus on one or two languages
 - Parallel and Distributed Databases
 - Object-based Databases
 - Blockchain
 - Advanced Query Processing and Optimization Techniques
 - Advanced Relational Algebra and Calculus
 - Advanced Data Mining and Analytics
 - Implement a Relational DB from Scratch

Data Definition and Manipulation

- Database languages
 - Data definition language
 - specify the database schema
 - Data manipulation language
 - express database queries and updates
- Data Definition Language (DDL)
 - DDL compiler generates a set of table templates stored in a data dictionary
 - Database schema (表头)
 - Integrity constraints
 - primary key, capacity > 0
 - <address, time> should be unique
 - Authorization (who can access it)

```
create table lab(  
    id serial primary key,  
    address varchar(20) not null,  
    time varchar(20) not null,  
    capacity int,  
    teacher varchar(20),  
    unique (address,time)  
);
```

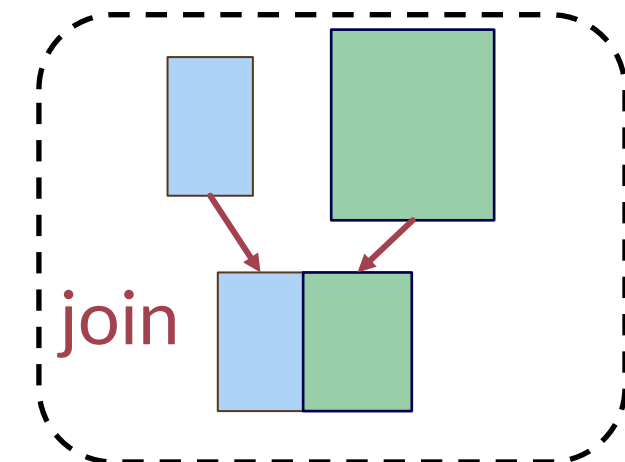
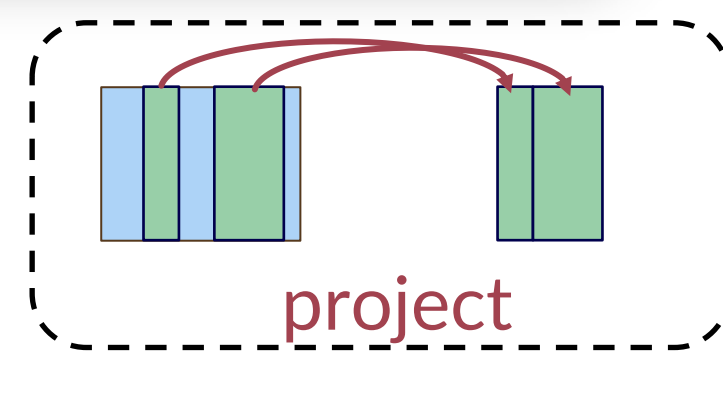
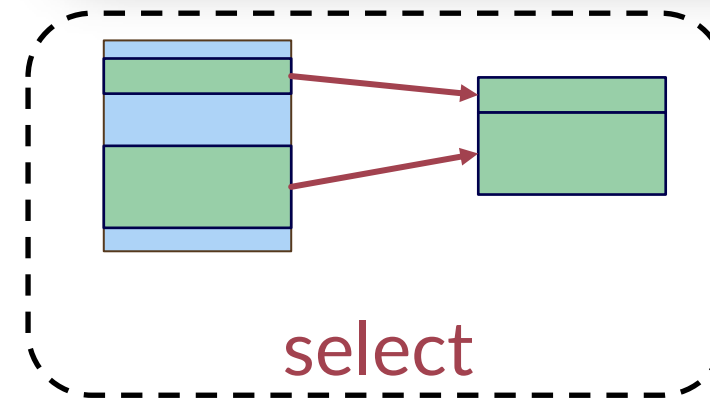
Data Definition and Manipulation

- Data Manipulation Language (DML)
 - **Language** for **accessing** and **updating** the **data** organized by the appropriate data model (also known as query language)
 - INSERT (增加)
 - DELETE (删除)
 - UPDATE (改动)
 - SELECT (查询)
- SQL (Structured Query Language)
 - Takes **several tables as input** (possibly only one) and always **returns a single table**



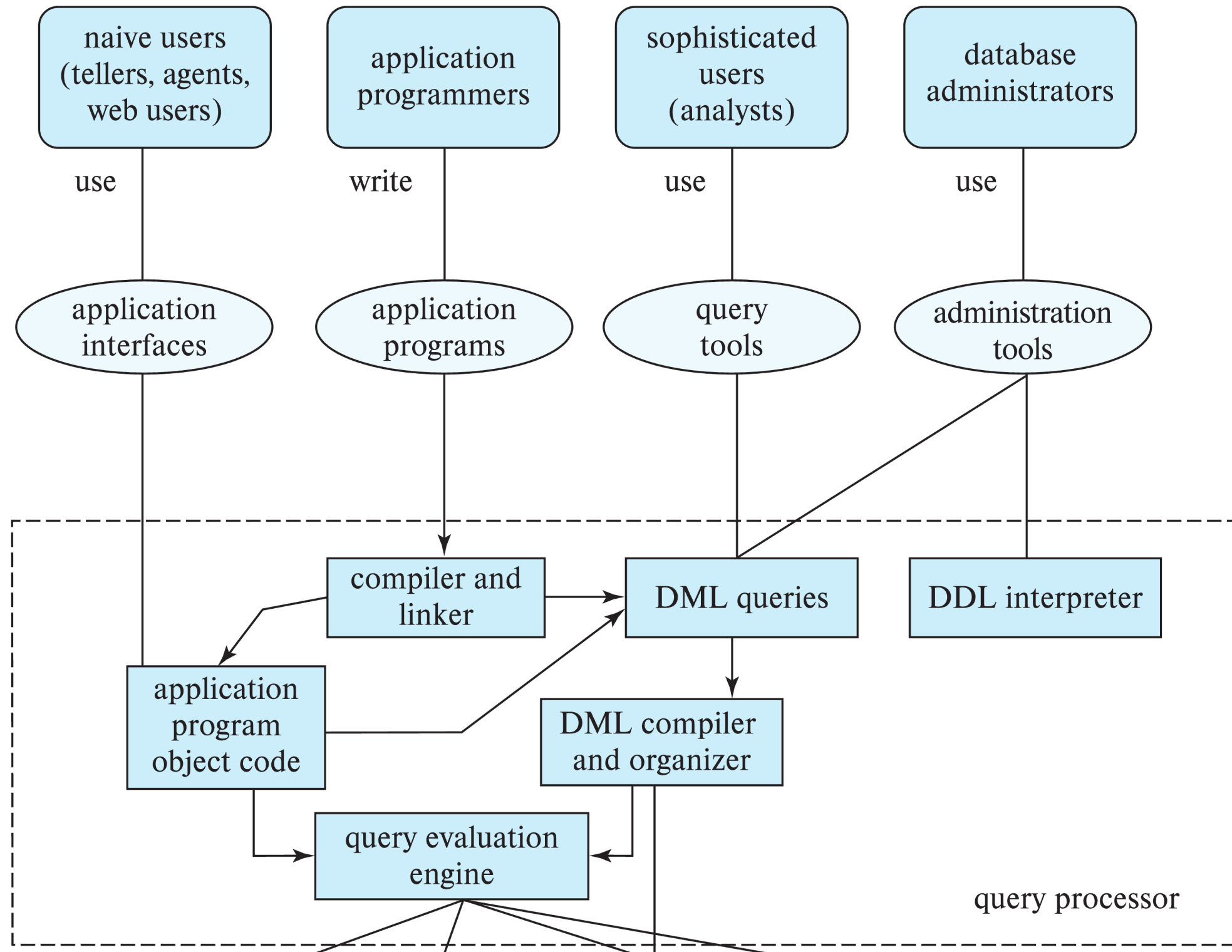
```
select * from lab;
```

```
select * from lab where time = '3-34';
```

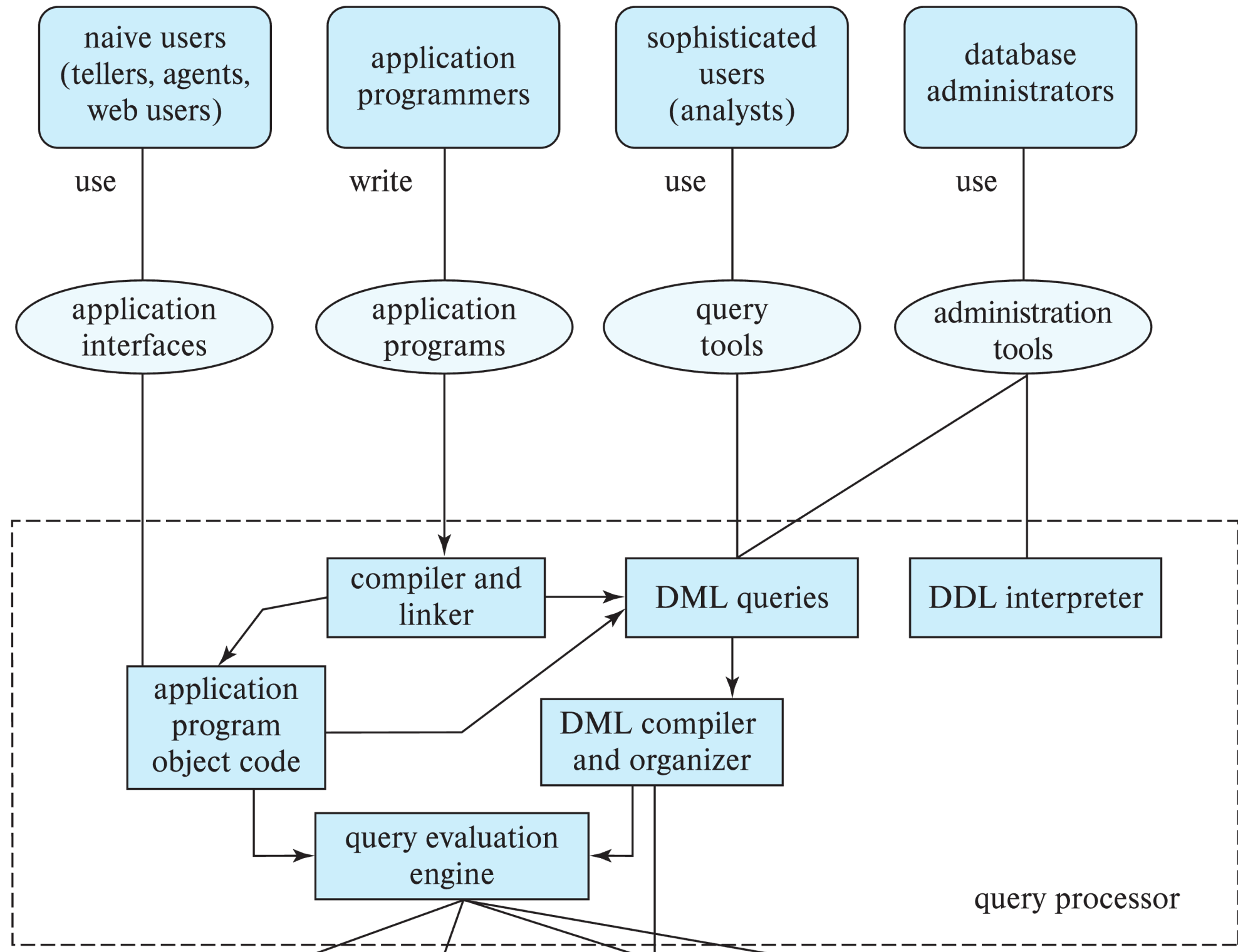


Database Users

- Naïve users
 - Use predefined user interfaces to interact
- Application programmers
 - Design user interfaces
- Sophisticated users
 - E.g., analyst using database query language and data analysis softwares
- Database admin
 - Schema definition
 - Storage structure and access-method definition
 - Schema and physical-organization modification
 - Granting of authorization for data access
 - Routine maintenance



Database Users



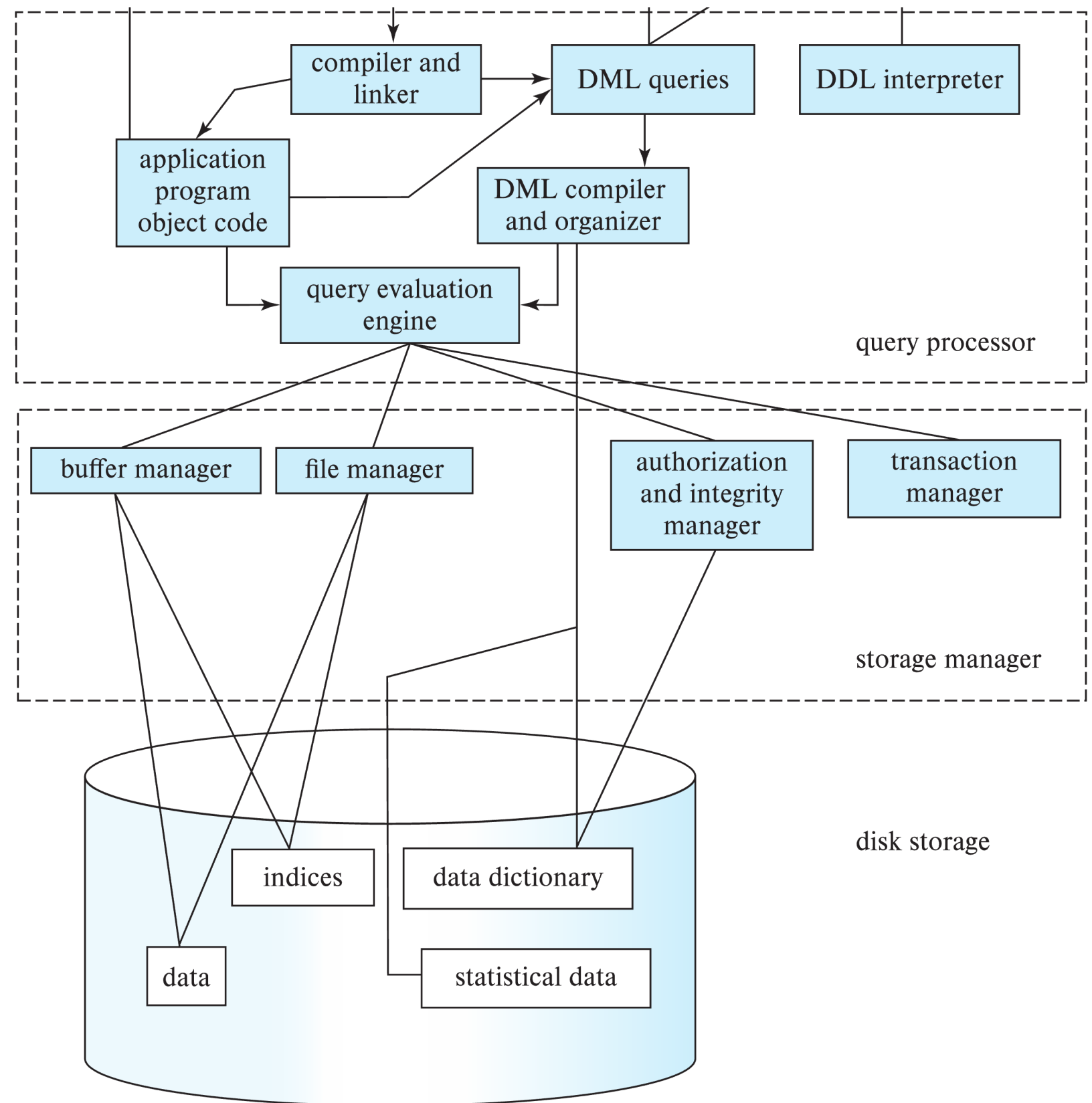
Query processor

- DDL interpreter
 - Record definitions of tables
- DML compiler
 - Translate DML code into low-level instructions for query evaluation engine
 - Query optimization
- Query evaluation engine
 - Run low-level instructions

Database Architecture

Storage manager

- Translates various DML statements into **low-level file-system commands**
- Responsible for **storing, retrieving, and updating** data in the database



Notices

- Find your own lab session (2 different groups)
 - Attendance is required
- WeCom Group
 - Teaching assistants will be there to help you
- (For international students) If you don't know how to join the WeCom group or Blackboard site, please email me or find TAs as soon as possible
 - My Email: wangzq3@sustech.edu.cn

**Please remember to attend your lab class.
It is strongly recommended to install the required software before your lab session.**